

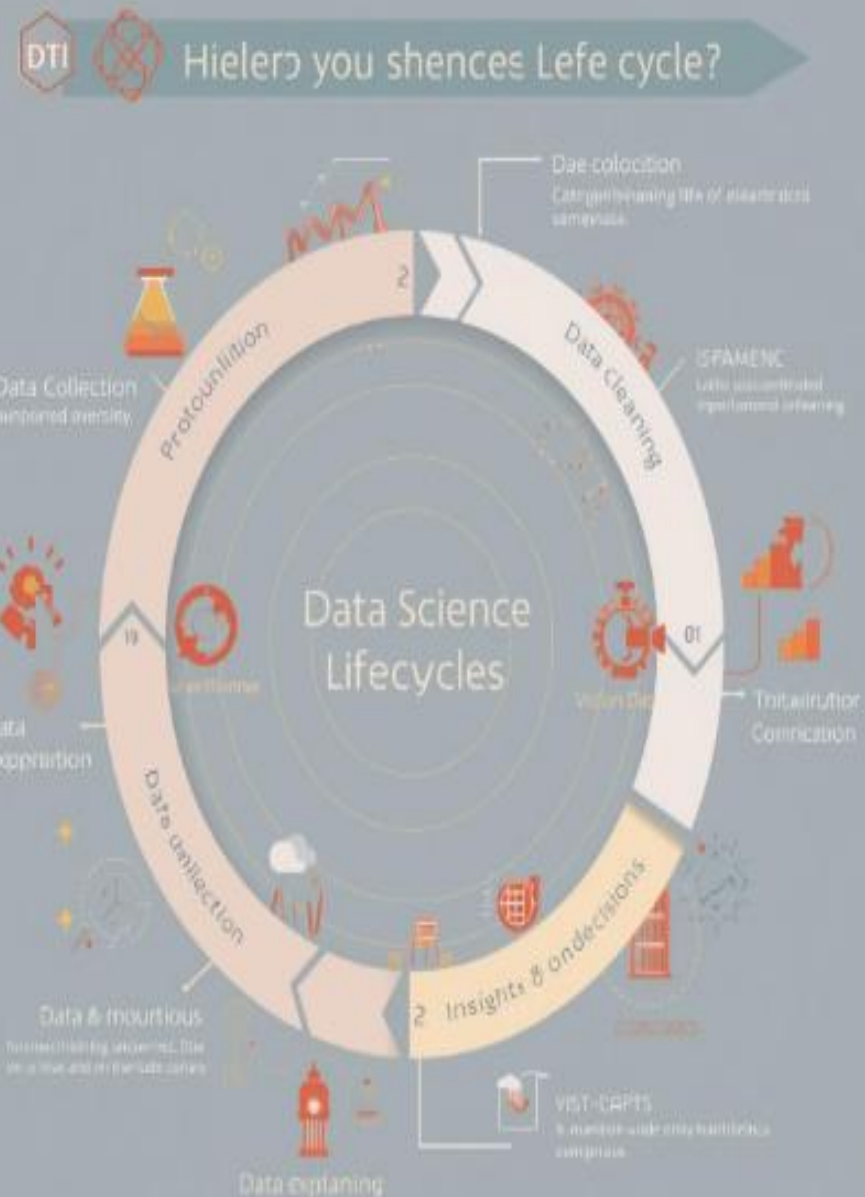
Impact of Screen Time

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Vijay



Data Science Lifecycle Overview

The data science process comprises essential steps: defining the problem, collecting and cleaning data, exploring and analyzing relationships, visualizing findings, and deriving actionable insights for effective decision-making.



Problem Definition and Data Collection

Lifestyle Changes Due to Excessive Screen Time



	158
	165
	165
155cm	
	165.1
	156
	160
	162
	182
	182
	165
	162
158cm	
	166
5.6 inch	
	185
	168
	170
	167.64
	165
	168
	186
	150
	155
	160
	155
	155
	155
	170
	162

	65
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	111
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48q	60
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	44
	55
	88
	65
	52
	70
	62.5

Importance of Data Cleaning

Data cleaning is crucial for obtaining **accurate results**. This step involves addressing missing values, removing inconsistencies, and standardizing formats to ensure the data's reliability before analysis begins.



Handle Missing Values

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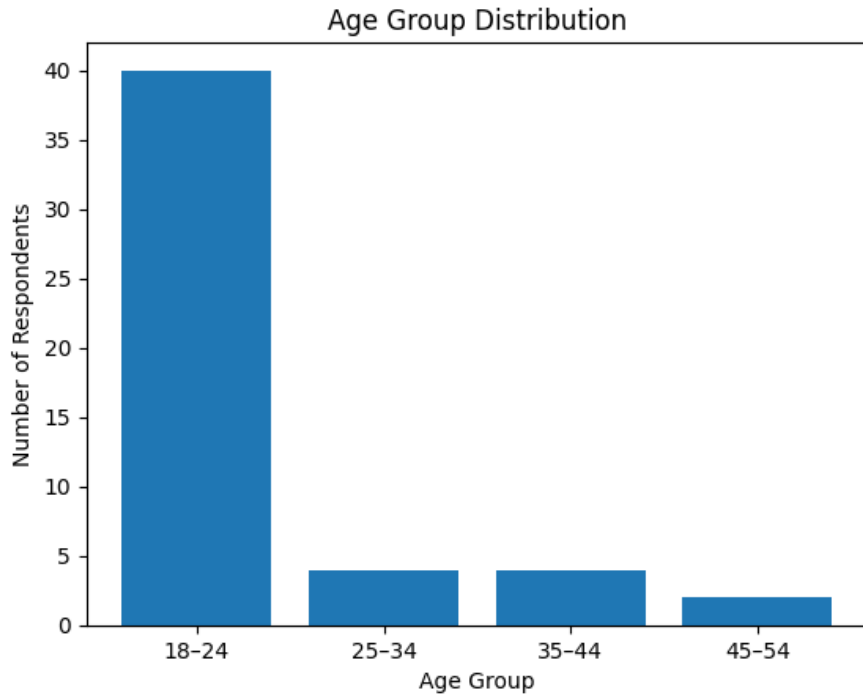
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Exploratory Data Analysis Insights

Exploratory Data Analysis (EDA) is crucial for understanding distributions and relationships among variables, detecting outliers, and forming hypotheses that shape subsequent analysis, ultimately ensuring a robust data-driven approach.



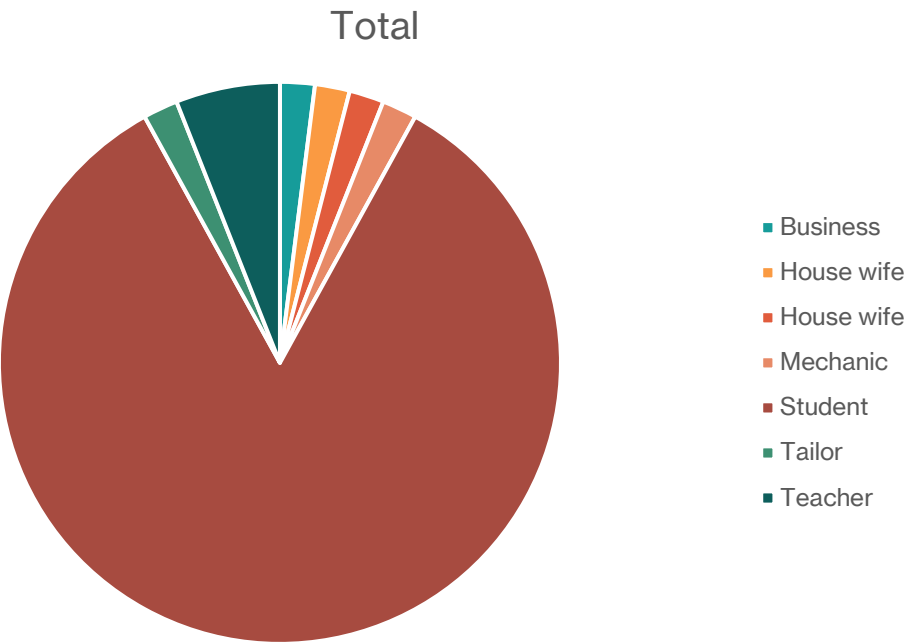


Age Group Distribution	Age Group	Number of Respondents
	18–24 years	40
	25–34 years	4
	35–44 years	4
	45–54 years	2
	Total	50

Age Group Distribution

Majority of respondents belong to the 18–24 years age group, indicating a youth-focused sample.

Occupation Distribution



Row Labels	Count of What is your primary occupation?
Business	1
House wife	1
House wife	1
Mechanic	1
Student	42
Tailor	1
Teacher	3
Grand Total	50

Mobile Screen Time Distribution

Number of Respondents



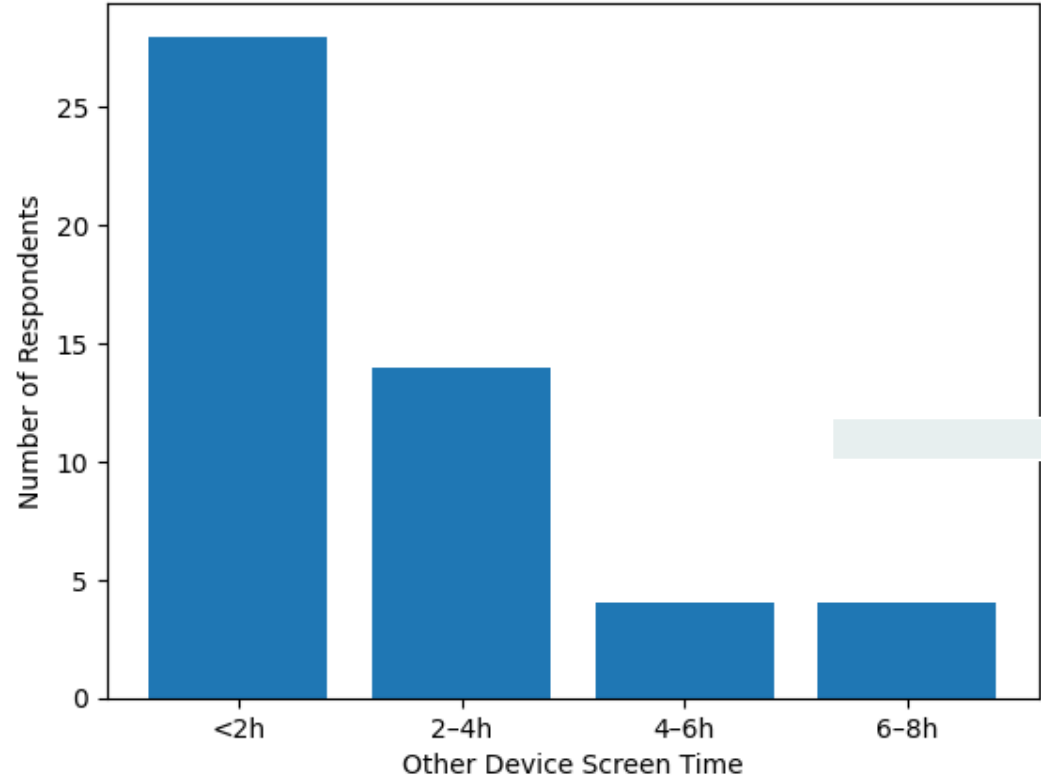
Most respondents spend between 2–6 hours daily on mobile phones for non-productive activities.

■ 18–24 years ■ 25–34 years ■ 35–44 years ■ 45–54 years ■ Total

Age Group Distribution	Age Group	Number of Respondents
	18–24 years	40
	25–34 years	4
	35–44 years	4
	45–54 years	2
	Total	50

Other Device Screen Time

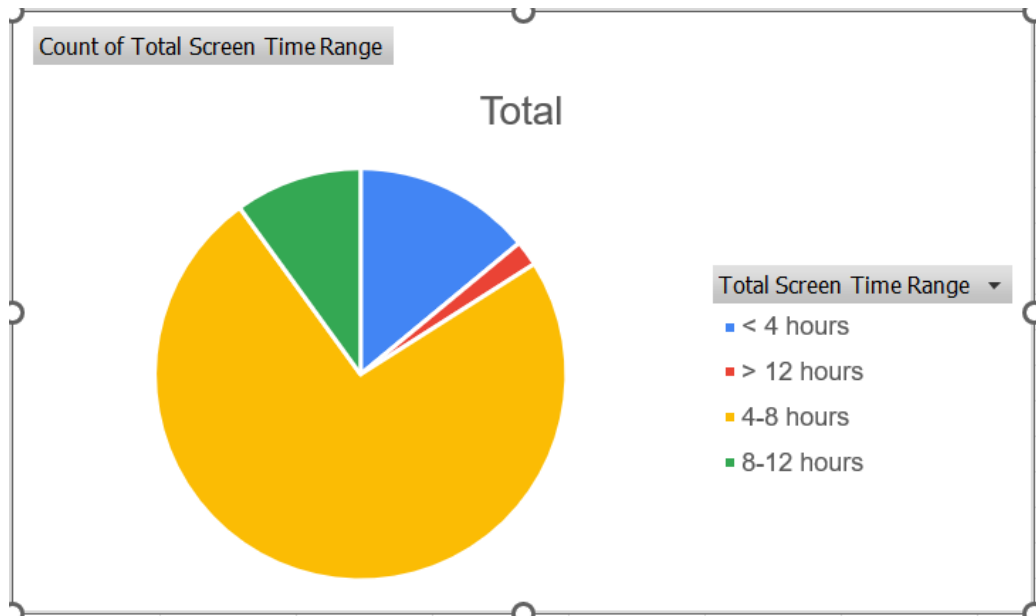
Other Device Screen Time Distribution



Non-mobile screen usage is mostly below 2 hours for a majority of respondents.

Other Device Screen Time Distribution	Other Device Screen Time	Number of Respondents
	< 2 hours	28
	2–4 hours	14
	4–6 hours	4
	6–8 hours	4
	Total	50

Total Screen Time Distribution

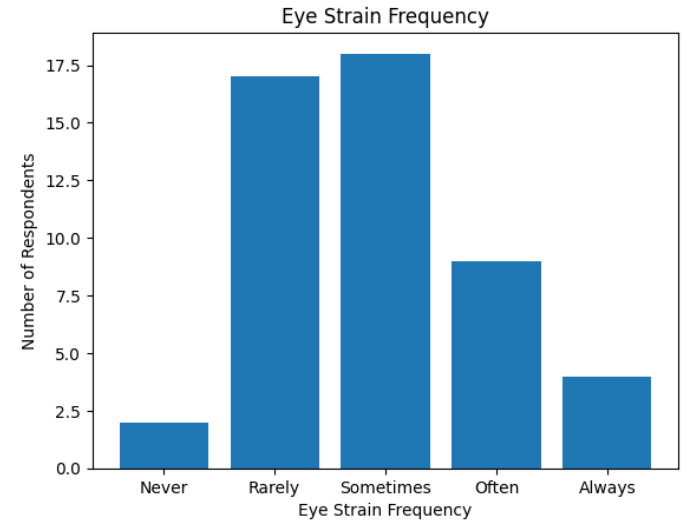


Most participants fall in the 4–8 hours total daily screen time category.

Total Screen Time Distribution	Total Screen Time	Number of Respondents
	< 4 hours	7
	4–8 hours	32
	8–12 hours	10
	> 12 hours	1
	Total	50

Eye strain symptoms increase significantly with prolonged screen exposure.

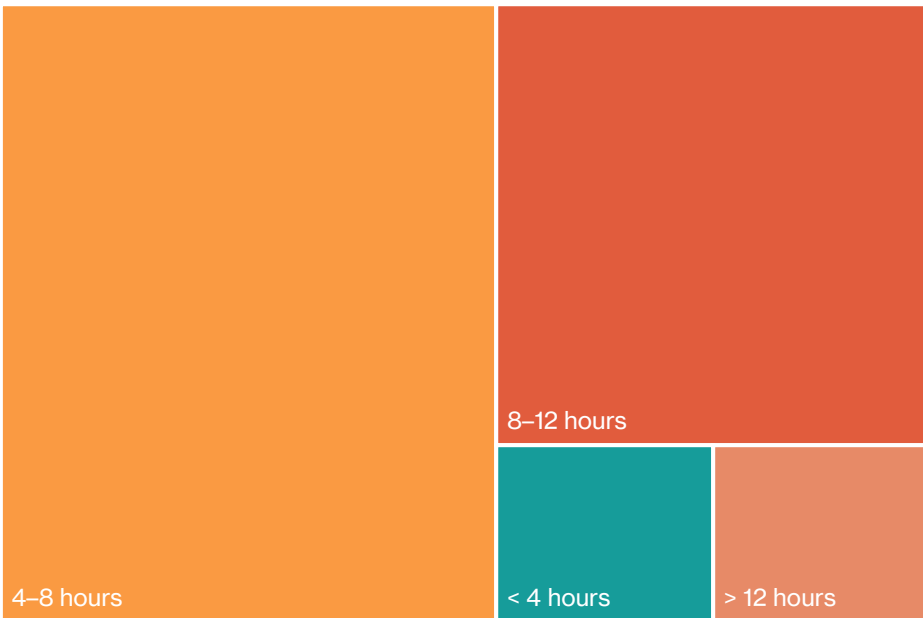
Eye Strain Frequency



Eye Strain Frequency	Eye Strain Frequency	Number of Respondents
	Never	2
	Rarely	17
	Sometimes	18
	Often	9
	Always	4
	Total	50

Chart Title

■ Often / Always Eye Strain



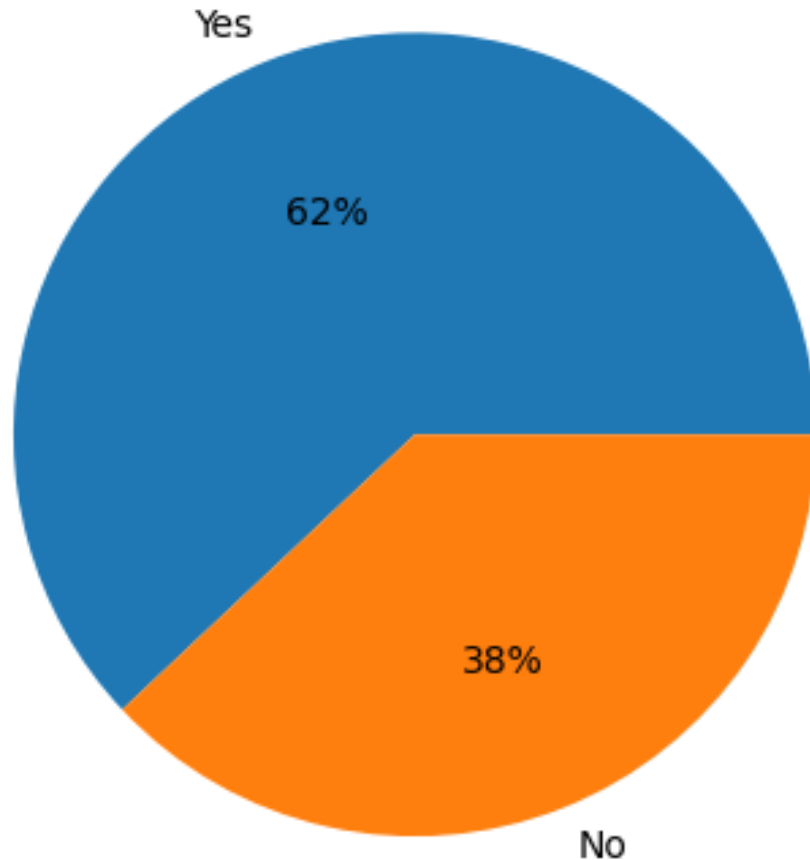
Screen Time vs Eye Strain	Total Screen Time	Often / Always Eye Strain
	< 4 hours	1
	4-8 hours	8
	8-12 hours	5
	> 12 hours	1

Screen Time vs Eye Strain

Eye strain (Often/Always) increases sharply with higher screen time categories.

Glasses / Contact Lens Usage

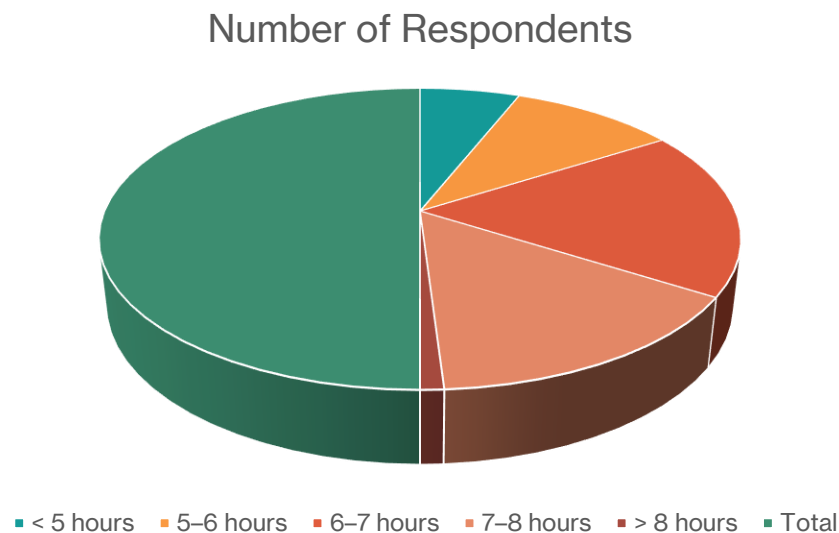
Glasses / Contact Lens Usage



More than half of the respondents use glasses or contact lenses.

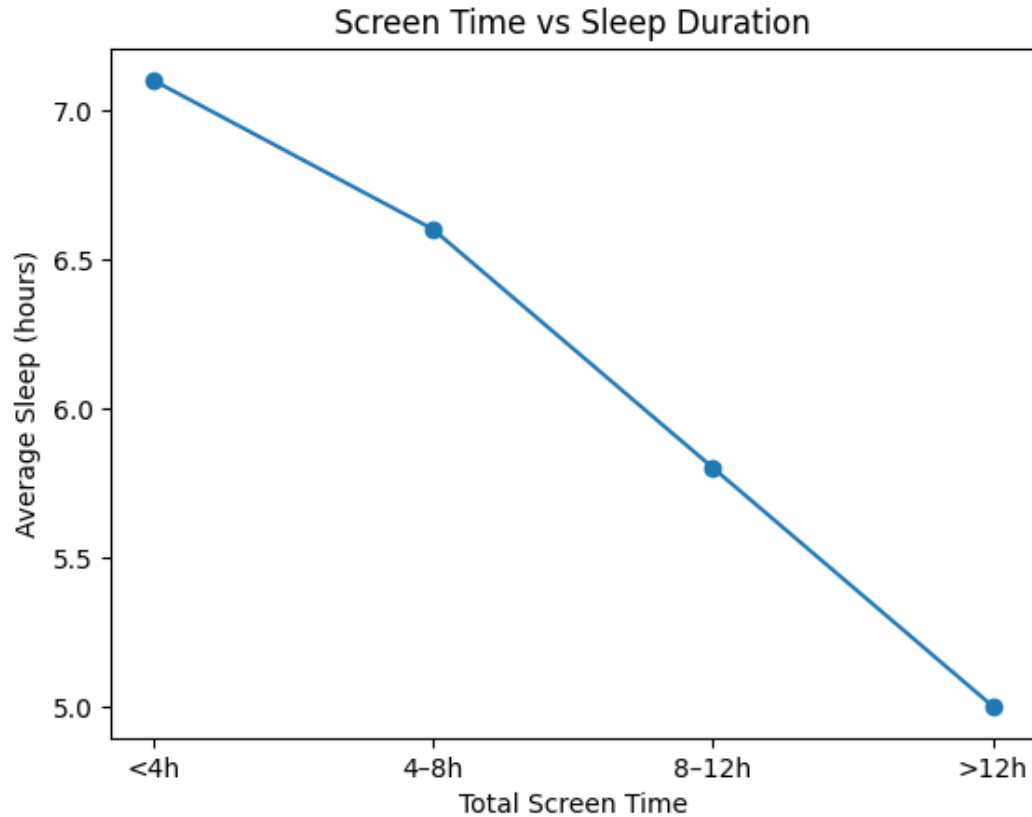
Glass / Lens Usage	Uses Glasses / Contact Lens	Number of Respondents
	Yes	31
	No	19
	Total	50

Sleep Duration Distribution



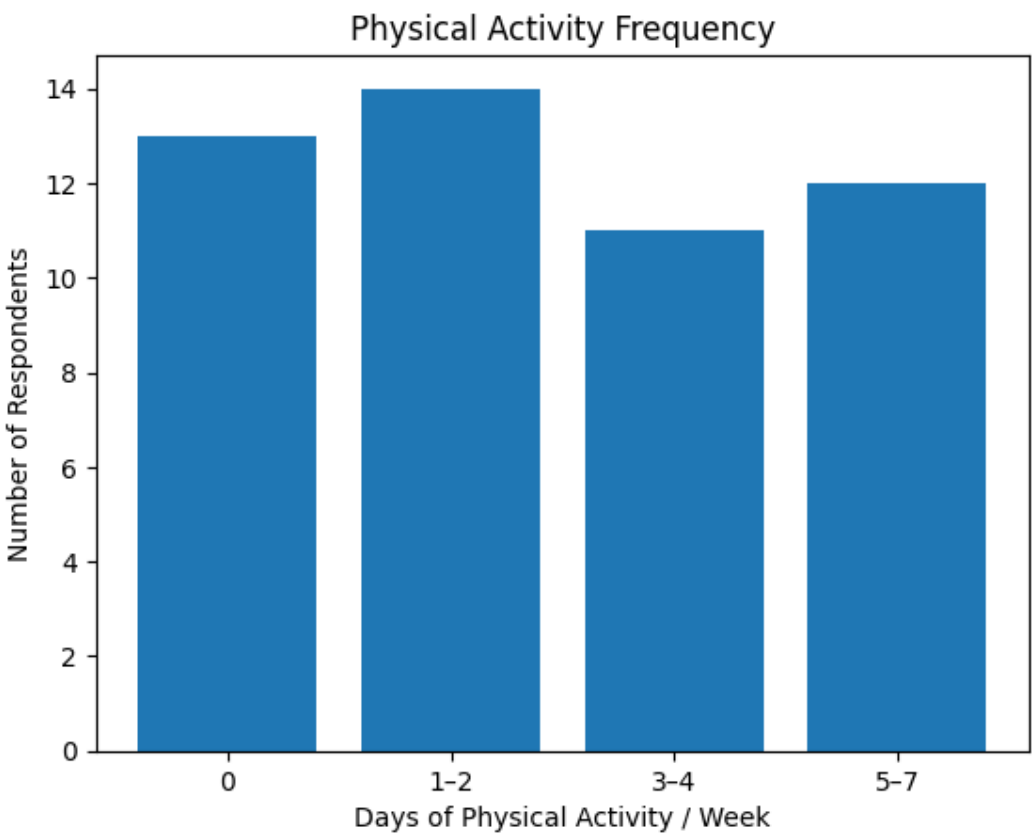
Sleep Duration Distribution	Sleep Duration	Number of Respondents
	< 5 hours	6
	5-6 hours	10
	6-7 hours	18
	7-8 hours	15
	> 8 hours	1
	Total	50

Screen Time vs Sleep Duration



Clear negative relationship: higher screen time results in lower sleep duration.

Physical Activity Frequency

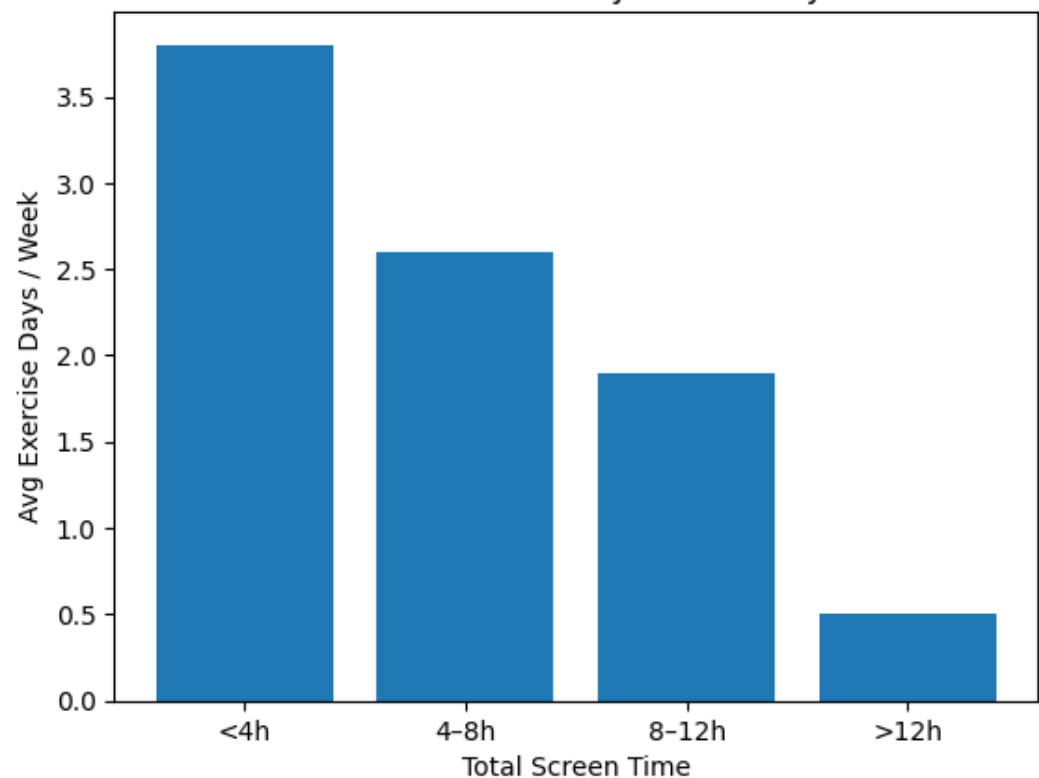


Nearly half of respondents engage in physical activity less than 3 days per week.

Physical Activity Frequency	Days of Physical Activity / Week	Number of Respondents
	0 days	13
	1-2 days	14
	3-4 days	11
	5-7 days	12
	Total	50

Screen Time vs Physical Activity

Screen Time vs Physical Activity

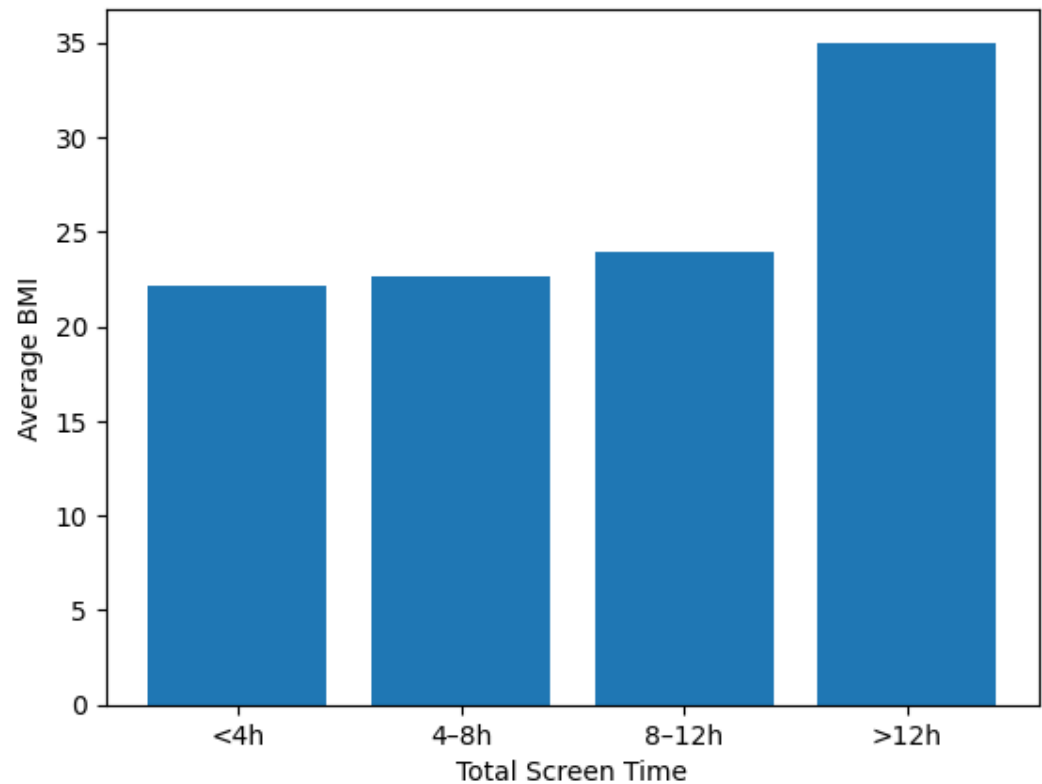


Higher screen time leads to reduced physical activity levels.

Screen Time vs Physical Activity	Total Screen Time	Avg Exercise Days / Week
	< 4 hours	3.8
	4–8 hours	2.6
	8–12 hours	1.9
	> 12 hours	0.5

Screen Time vs BMI

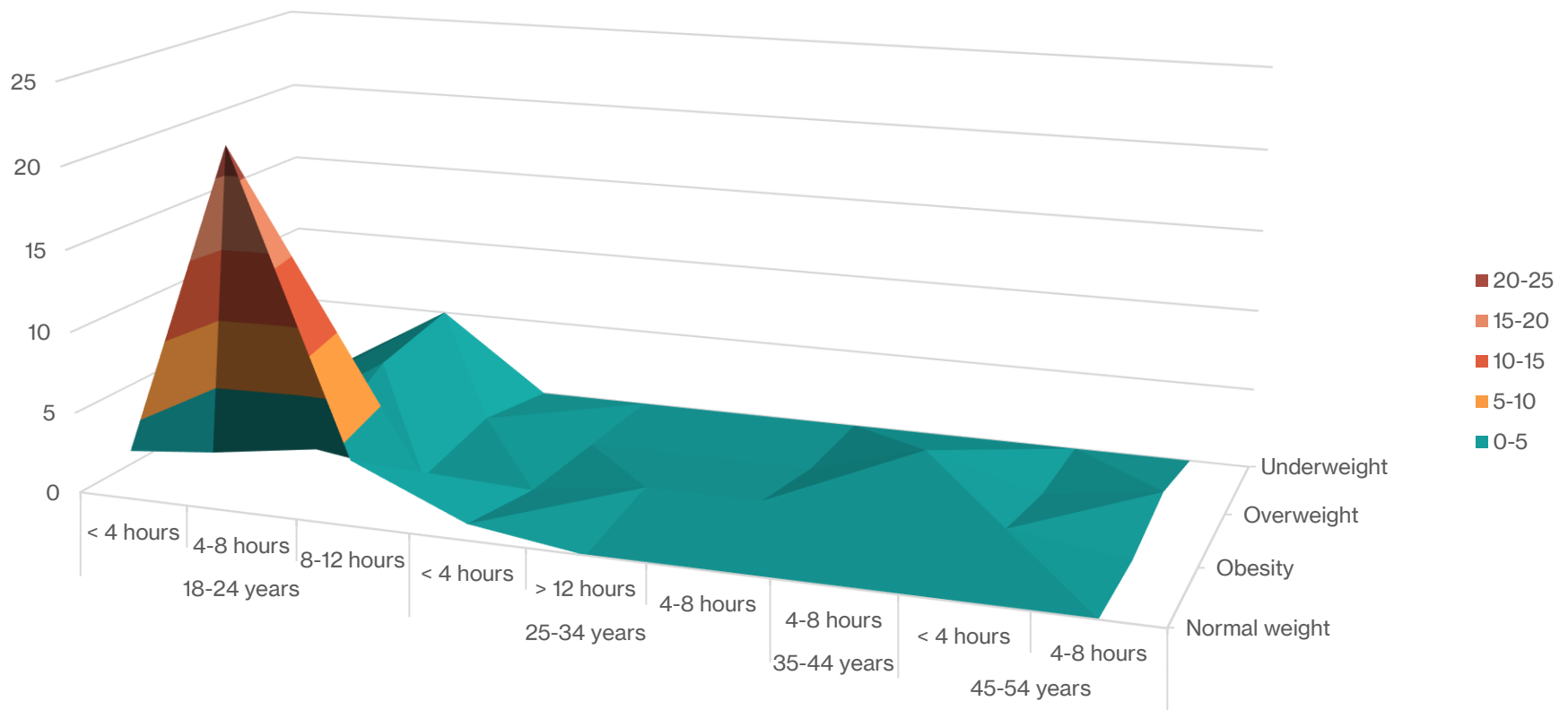
Screen Time vs BMI



Average BMI increases with screen time, indicating obesity risk.

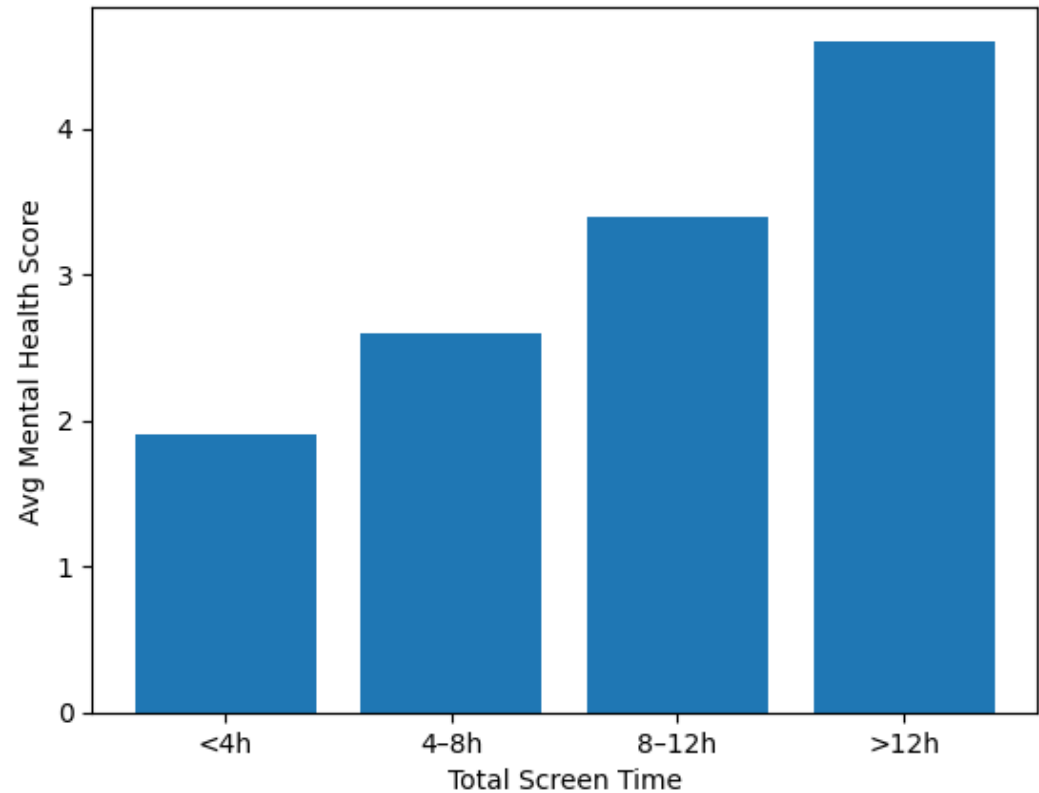
Screen Time vs BMI	Total Screen Time	Average BMI
	< 4 hours	22.1
	4–8 hours	22.6
	8–12 hours	23.9
	> 12 hours	35

Screen Time vs BMI vs Age group



Screen Time vs Mental Health

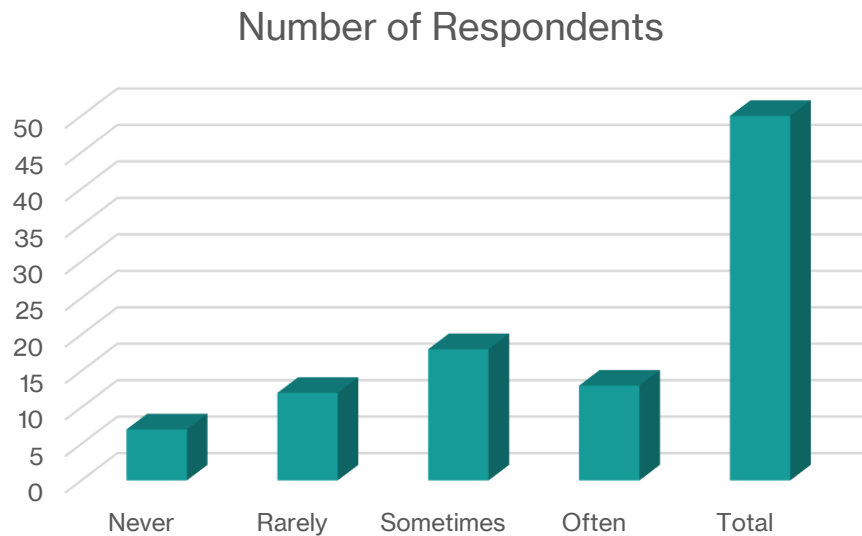
Screen Time vs Mental Health Score



Mental health scores worsen as screen time increases.

Screen Time vs Mental Health Score	Total Screen Time	Avg Mental Health Score
	< 4 hours	1.9
	4–8 hours	2.6
	8–12 hours	3.4
	> 12 hours	4.6

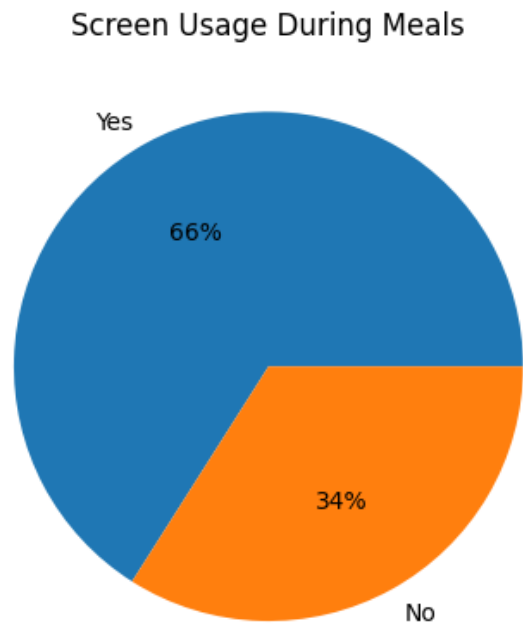
Pain due to screen time



Back / Neck / Shoulder Pain Due to Screen	Pain Frequency	Number of Respondents
	Never	7
	Rarely	12
	Sometimes	18
	Often	13
	Total	50

Screen Usage During Meals

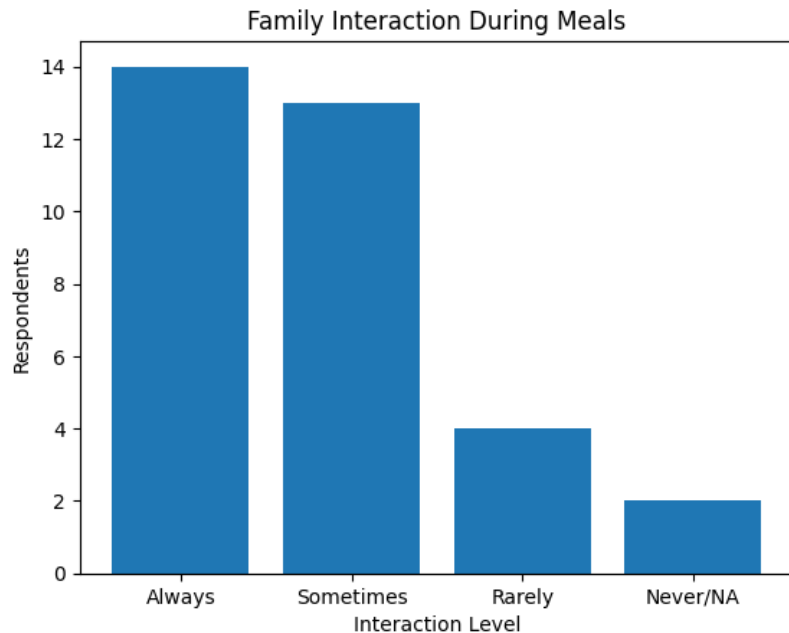
A large portion of respondents use screens during meals.



Screen Usage During Meals	Screen During Meals	Number of Respondents
	Yes	33
	No	17
	Total	50

Family Interaction During Meals

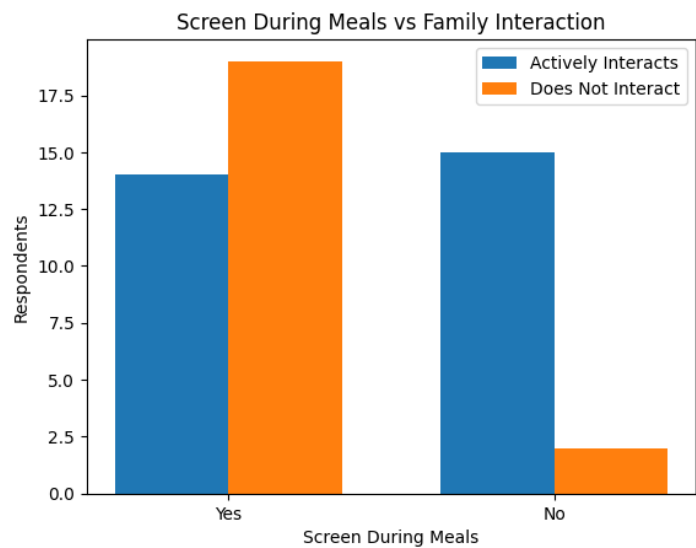
Screen usage during meals reduces family interaction.



Family Interaction During Meals	Interaction Level	Number of Respondents
	Always	14
	Sometimes	13
	Rarely	4
	Never / Not applicable	2

Screen During Meals vs Family Interaction

Respondents not using screens during meals interact more actively with family.



SCREEN DURING MEALS VS FAMILY INTERACTION	SCREEN DURING MEALS	ACTIVELY INTERACTS	DOES NOT INTERACT
	Yes	14	19
	No	15	2

Conclusion & Recommendations

